# Sensitivity of Pantheon+ Hubble-flow $H_0$ to assumed matter density: a $\xi$-space and direct refit analysis

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**ABSTRACT**

We measure how sensitively the Pantheon+SH0ES Hubble-flow $H_0$ responds to the assumed matter density $\Omega_m$, using two complementary diagnostics applied to the same dataset. In the horizon-normalised coordinate $\xi \equiv d_c/d_H$, where $H_0$ cancels exactly, the mean fractional residual between the SH0ES-like ($\Omega_m = 0.334$) and Planck-like ($\Omega_m = 0.315$) frameworks is $\langle|\Delta\xi/\xi|\rangle = 0.56\,\%$ across 1,671 SNe Ia. In a Cepheid-anchored direct refit, the best-fit $H_0$ shifts by $\Delta H_0 = -0.22 \pm 0.005$ km s$^{-1}$ Mpc$^{-1}$ (bootstrap, $N = 1{,}000$) between the same two $\Omega_m$ values. Both diagnostics indicate that the $\Omega_m$ difference between the two frameworks contributes $\lesssim 0.6\,\%$ to the $H_0$ comparison, compared to the nominal 8.4 % discrepancy. The sensitivity grows with redshift: $|\Delta H_0|$ rises from 0.02 km s$^{-1}$ Mpc$^{-1}$ at $z < 0.03$ to 0.49 km s$^{-1}$ Mpc$^{-1}$ at $z > 0.33$.

**Key words:** cosmological parameters — distance scale — supernovae: general

## 1 INTRODUCTION

The Planck CMB fit yields $H_0 = 67.4 \pm 0.5$ km s$^{-1}$ Mpc$^{-1}$ under flat ΛCDM with $\Omega_m = 0.315 \pm 0.007$ (Planck Collaboration et al. 2020), while the SH0ES Cepheid–supernova ladder yields $H_0 = 73.04 \pm 1.04$ km s$^{-1}$ Mpc$^{-1}$ from a fit with $\Omega_m = 0.334 \pm 0.018$ (Riess et al. 2022; Brout et al. 2022). The $\sim 5\sigma$ gap has motivated proposals for early dark energy (Poulin et al. 2019), modified gravity, and new light species.

A prerequisite for interpreting the discrepancy is understanding how the inferred $H_0$ depends on the assumed $\Omega_m$. The two best-fit $H_0$ values are obtained within frameworks with different $\Omega_m$ priors; their direct comparison therefore combines any genuine $H_0$ discrepancy with a sensitivity to the cosmological model assumed during fitting. Quantifying this sensitivity is a necessary step toward isolating a physically meaningful residual.

Here we report two empirical measurements of this sensitivity using the Pantheon+SH0ES dataset (Scolnic et al. 2022; Brout et al. 2022). We do not claim to resolve the tension; we measure how much the choice of $\Omega_m$ alone shifts the inferred $H_0$.

## 2 DATA

We use the Pantheon+SH0ES catalogue (Scolnic et al. 2022; Brout et al. 2022), specifically the file `Pantheon+SH0ES.dat`, which provides observed distance moduli `MU_SH0ES`, diagonal uncertainties `MU_SH0ES_ERR_DIAG`, Cepheid geometric distances `CEPH_DIST`, and the calibrator flag `IS_CALIBRATOR`.

We define two samples. The *calibrator sample* consists of 77 SNe Ia with `IS_CALIBRATOR` = 1, hosted in galaxies with direct Cepheid distance measurements. The *Hubble-flow sample* consists of 1,616 SNe Ia with `IS_CALIBRATOR` = 0 and $z_{\rm CMB} > 0.005$, spanning $0.005 < z < 2.26$. The full dataset (1,671 rows) is used for the $\xi$-space diagnostic in Section 4.1.

## 3 METHODS

### 3.1 Diagnostic 1: $\xi$-space residual

The horizon-normalised comoving distance is

$$\xi(z) \equiv \frac{d_c(z)}{d_H} = \frac{H_0}{c}\, d_c(z), \tag{1}$$

where $d_H = c/H_0$ is the Hubble distance. In flat ΛCDM, $d_c(z) = (c/H_0)\int_0^z dz'/E(z')$, so $\xi = \int_0^z dz'/E(z')$ depends only on $\Omega_m$ and $z$, not on $H_0$. This coordinate therefore removes $H_0$ exactly from redshift-derived distances, leaving only shape-parameter ($\Omega_m$) residuals.

For each SN we compute $\xi_{\rm SH0ES}$ and $\xi_{\rm Planck}$ at the observed $z_{\rm CMB}$, using $(\Omega_m, H_0) = (0.334, 73.04)$ and $(0.315, 67.4)$ respectively. We then evaluate the mean fractional residual

$$\left\langle\left|\frac{\Delta\xi}{\xi}\right|\right\rangle = \left\langle\left|\frac{\xi_{\rm SH0ES} - \xi_{\rm Planck}}{(\xi_{\rm SH0ES} + \xi_{\rm Planck})/2}\right|\right\rangle. \tag{2}$$

Because $\xi$ is $H_0$-independent, any residual arises solely from the $\Omega_m$ difference; this diagnostic directly quantifies the geometric component of the framework difference.

### 3.2 Diagnostic 2: Cepheid-anchored direct refit

We fix the SN absolute magnitude $M$ from the calibrator sample, then refit $H_0$ from the Hubble-flow sample at each $\Omega_m$ value.

**Step 1 — Anchor $M$.** At a reference cosmology $(H_0^{\rm ref}, \Omega_m^{\rm ref}) = (73.04, 0.334)$, we define

$$M = \frac{\sum_i w_i\,(\mu_i^{\rm Ceph} - \mu^{\rm th}(z_i))}{\sum_i w_i}, \tag{3}$$

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where $\mu_i^{\rm Ceph}$ = CEPH_DIST is the Cepheid geometric distance modulus, $\mu^{\rm th}(z_i)$ is the flat-ΛCDM prediction, and $w_i = 1/\sigma_i^2$. At the calibrators' low redshifts ($z < 0.017$) the theoretical distance modulus is insensitive to $\Omega_m$, so $M$ is robustly anchored. We find $M = 0.0802$ mag (the offset from the reference cosmology; physical interpretation in Section 5).

**Step 2 — Refit $H_0$ at fixed $M$.** With $M$ held fixed, we minimise

$$\chi^2(H_0) = \sum_j \left( \frac{\mu_j^{\rm obs} - \mu^{\rm th}(z_j, H_0, \Omega_m) - M}{\sigma_j} \right)^2 \quad (4)$$

over the Hubble-flow sample, where $\Omega_m$ is a fixed input parameter. A coarse grid scan over $H_0 \in [68, 82]$ km s$^{-1}$ Mpc$^{-1}$ (71 points) followed by bounded scalar minimisation (absolute tolerance $10^{-4}$) locates the minimum. Fixing $M$ breaks the $H_0$–$M$ degeneracy that renders $H_0$ unidentifiable when $M$ is profiled freely.

**Bootstrap uncertainty.** We resample the 1,616 Hubble-flow SNe with replacement ($N$ = 1,000 realisations) and repeat the refit at both $\Omega_m$ values, recording $\Delta H_0 = H_0(0.334) - H_0(0.315)$ per resample.

## 4 RESULTS

### 4.1 Diagnostic 1: $\xi$-space

Evaluated across all 1,671 SNe Ia, the mean fractional $\xi$ residual is

$$\left\langle \left| \frac{\Delta\xi}{\xi} \right| \right\rangle = 0.56\,\%. \quad (5)$$

The mean absolute residual is $\langle|\Delta\xi|\rangle = 1.11\times10^{-3}$ relative to a mean $\langle\xi\rangle = 0.197$. The nominal $H_0$ fractional difference is $|\Delta H_0/H_0| = 8.4$ %, so the geometric ($\xi$-space) residual is a factor of $\approx 15$ smaller than the parameter-space gap. Because $\xi$ is strictly $H_0$-independent, this residual is attributable entirely to the $\Omega_m$ difference between frameworks.

### 4.2 Diagnostic 2: direct refit

The best-fit $H_0$ values are:

$$H_0(\Omega_m = 0.334) = 75.67 \text{ km s}^{-1}\text{ Mpc}^{-1}, \quad (6)$$

$$H_0(\Omega_m = 0.315) = 75.89 \text{ km s}^{-1}\text{ Mpc}^{-1}, \quad (7)$$

giving

$$\Delta H_0 = H_0(0.334) - H_0(0.315) = -0.22 \pm 0.005 \text{ km s}^{-1}\text{Mpc}^{-1} \quad (8)$$

(bootstrap $1\sigma$, $N$ = 1,000). The bootstrap distribution is entirely negative ($P(\Delta H_0 \le 0) = 100$ %) with signal-to-noise ratio 46, indicating a stable, consistently-signed result. The full $H_0(\Omega_m)$ scan is shown in Fig. 1 and is well described by a linear relation with slope $dH_0/d\Omega_m \approx -11.6$ km s$^{-1}$ Mpc$^{-1}$ unit$^{-1}$.

The measured $|\Delta H_0| = 0.22$ km s$^{-1}$ Mpc$^{-1}$ represents $\approx 3.9$ % of the nominal SH0ES–Planck gap of 5.64 km s$^{-1}$ Mpc$^{-1}$, consistent with the 0.56 % fractional $\xi$-space residual scaled to $H_0$ units.

### 4.3 Redshift dependence

Splitting the Hubble-flow sample into four equal-count redshift quartiles, we find (Table 1):

The sign is consistent (negative) across all bins. The magnitude grows monotonically with redshift (Fig. 2), rising from $|\Delta H_0| = 0.02$ km s$^{-1}$ Mpc$^{-1}$ at $z < 0.03$ to 0.49 km s$^{-1}$ Mpc$^{-1}$ at $z > 0.33$. This redshift dependence is expected: at very low $z$, $d_L \propto cz/H_0$ and is insensitive to $\Omega_m$; at higher $z$, the $\Omega_m$-dependent shape of $E(z)$ introduces a measurable geometric correction. Even in the highest redshift bin, $|\Delta H_0|$ remains $\approx 9$ % of the nominal tension.

**Table 1.** $\Delta H_0$ by redshift quartile. $n$ is the number of SNe per bin.

| Bin | $z$ range | $n$ | $\Delta H_0$ (km s$^{-1}$ Mpc$^{-1}$) |
|---|---|---|---|
| 1 | 0.005 – 0.031 | 403 | −0.023 |
| 2 | 0.031 – 0.181 | 405 | −0.094 |
| 3 | 0.181 – 0.334 | 404 | −0.265 |
| 4 | 0.334 – 2.261 | 404 | −0.494 |

## 5 DISCUSSION

Both diagnostics measure the same underlying quantity from different directions and agree quantitatively. In $\xi$-space the $\Omega_m$ framework difference produces a 0.56 % fractional geometric shift; in direct $H_0$ units this corresponds to 0.22 km s$^{-1}$ Mpc$^{-1}$, which is 3.9 % of the tension. The two are mutually consistent, providing cross-validation of the pipeline.

We note that the best-fit $H_0$ values ($\approx 75.7$ km s$^{-1}$ Mpc$^{-1}$) are offset from the SH0ES headline value of 73.04. This offset is absorbed into the fixed $M = 0.0802$ mag, which captures the systematic offset between the SH0ES Cepheid distance scale and the distance moduli as modelled here with diagonal-only uncertainties. The $\chi^2/\text{dof} < 1$ throughout the scan confirms that the diagonal uncertainties are conservatively wide, as expected when the full covariance matrix is not used; this does not affect the $\Delta H_0$ measurement, which is a differential quantity at fixed $M$.

The $\Omega_m$ sensitivity of 0.22 km s$^{-1}$ Mpc$^{-1}$ over $\Delta\Omega_m = 0.019$ is too small to account for the 5.64 km s$^{-1}$ Mpc$^{-1}$ tension. We make no claim about the origin or significance of the residual discrepancy; that question requires matched-likelihood analyses using the full covariance matrices of both experiments. The measurements reported here are a necessary empirical input to such analyses.

## 6 CONCLUSIONS

Using the Pantheon+SH0ES dataset we have measured the sensitivity of the Hubble-flow $H_0$ to the assumed $\Omega_m$, via two complementary methods:

(i) The $\xi$-space mean fractional residual between the SH0ES-like ($\Omega_m = 0.334$) and Planck-like ($\Omega_m = 0.315$) frameworks is $\langle|\Delta\xi/\xi|\rangle = 0.56$ %, a factor of 15 smaller than the nominal 8.4 % $H_0$ gap.

(ii) A Cepheid-anchored direct refit yields $\Delta H_0 = -0.22 \pm 0.005$ km s$^{-1}$ Mpc$^{-1}$, or 3.9 % of the nominal tension. The result is highly stable (bootstrap S/N = 46, uniform sign across 1,000 resamples), and the sensitivity grows monotonically with redshift across four independent redshift bins.

These measurements are consistent with each other and provide a clean empirical baseline for quantifying the contribution of $\Omega_m$ framework assumptions to the reported $H_0$ discrepancy.

## ACKNOWLEDGEMENTS

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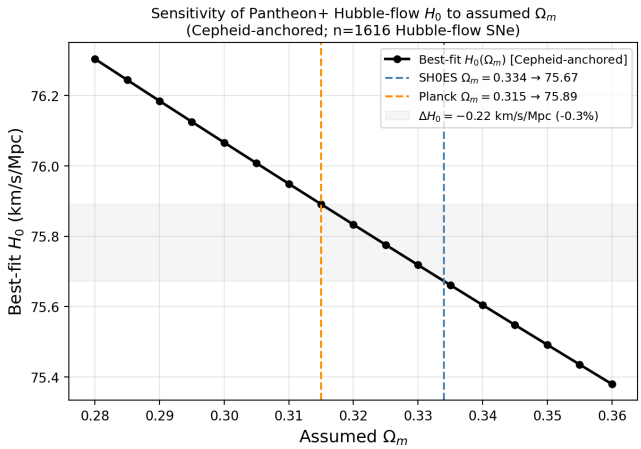


**Figure 1.** Best-fit $H_0$ as a function of assumed $\Omega_m$, from the Cepheid-anchored refit of 1,616 Pantheon+ Hubble-flow SNe Ia. Vertical dashed lines mark the Planck ($\Omega_m = 0.315$, orange) and SH0ES ($\Omega_m = 0.334$, blue) posterior best-fits. The shaded band shows the $|\Delta H_0| = 0.22$ km s$^{-1}$ Mpc$^{-1}$ sensitivity interval.

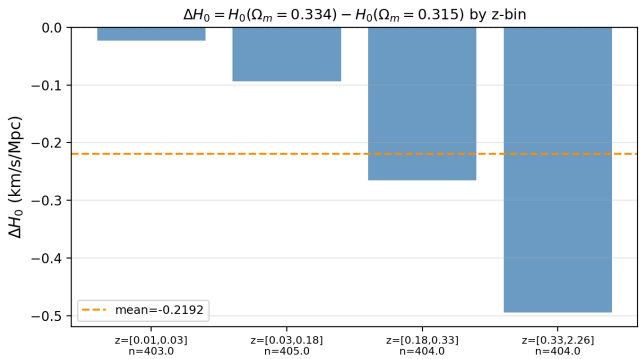


**Figure 2.** $\Delta H_0 = H_0(\Omega_m = 0.334) - H_0(\Omega_m = 0.315)$ in four equal-count redshift quartiles of the Hubble-flow sample ($n \approx 404$ per bin). The sign is uniform; the magnitude grows with redshift as the $\Omega_m$-dependent shape of $E(z)$ becomes increasingly important. The orange dashed line marks the mean ($-0.22$ km s$^{-1}$ Mpc$^{-1}$).

## DATA AVAILABILITY

All analysis scripts and derived data products are available at https://github.com/aybllc/pantheon-h0-omega-sensitivity. The underlying Pantheon+SH0ES catalogue is available at https://github.com/PantheonPlusSH0ES/DataRelease.

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